

MATLAB REVIEW

HANCHEN LI

Help

help	help	help browser	doc
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Housekeeping

clear window	clc	clear workspace	clear	clear graph	close
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Input and output

input numbers	=input(' ')	input text	=input(' ','s')	output	disp()
reading file	load('name.format')				

Type of value

8-bit integer	with symbol	int8()	translate number to array	num2str()
	without symbol	Uint8()		

Basic variables

π	pi	∞	inf	non-numerical	NaN
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Comparison

=	==	$\neq$	~=	>	>
$\geq$	>=	<	<	$\leq$	<=
not	~condition1	and	(con1)&(con2)	or	(con1) (con2)
xor=one true one false	xor(con1,con2)	all in array	all(condition)	any in array	any(condition)

Calculation

=	=	+	+	-	-
x	*	$\div$	/	a^n	$a^{(n)}$
e	exp()	ln	log()	$n!=1 \times 2 \times \dots \times n$	factorial(n)
sin	sin()	cos	cos()	tan	tan()
random between 0 to 1	rand()	remainder after $\div N$	mod(,N)	round to nearest	round()
round up	ceil()	round down	floor()	delete decimals	fix()

Arrays and sub-arrays

$(1 \ 2 \ 3)$	[1 2 3]	$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$	[1 ; 2 ; 3]	$\begin{pmatrix} 1 & 30 \\ 2 & 20 \\ 3 & 10 \end{pmatrix}$	[1 30 ; 2 20 ; 3 10]
$(a \ a+1 \ a+2 \ \dots \ b)$	a:b	$(a \ a+i \ a+2i \ \dots \ b)$	a:i:b	acegikm	'a':2:'m'
a×b all-zero matrix	zeros(a,b)	a×b all-one matrix	ones(a,b)	a×b ??? matrix	command(a,b)

extract 3 from $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$	A(7) or A(1,3)	extract $(1 \ 2 \ 3)$ from $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$	A(1 ; :)
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extract $\begin{pmatrix} 2 & 3 \\ 5 & 6 \end{pmatrix}$ from $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$	$A(1:2,2:3)$	extract $\begin{pmatrix} 1 & 3 \\ 7 & 9 \end{pmatrix}$ from $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$	$A(1:2:3 ; [1,3])$
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find maximum in A	in row	$\max(A, [], 2)$	find minimum A	in row	$\min(A, [], 2)$
	in column	$\max(A, [], 1)$		in column	$\min(A, [], 1)$
sum of A	in row	$\text{sum}(A, 2)$	average of A	in row	$\text{mean}(A, [], 2)$
	in column	$\text{sum}(A, 1)$		in column	$\text{mean}(A, [], 1)$
arrange A	from min to max	$\text{sort}(A, 1 \text{ or } 2)$	arrange A according to the N th column		$\text{sortrows}(A, N, ' ')$
	from max to min	$\text{sort}(A, 1 \text{ or } 2, 'descend')$			

find elements in A that meet condition	$\text{found} = \text{find}(A(\text{condition}))$	swap elements with other value in A	$A(\text{found}) = \text{value}$
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Array calculation

=	=	A+b or A+B	+	A-b or A-B	-
$A \times B$	*	A/B	/	A^b	^
$\times$	.*	$\div$	./	A^N	$A.^N$
if $Y \times X = 1$	$Y = \text{inv}(X) = X^{-1}$	swap row & column	$A.'$ or $\text{transpose}(A)$	/ and \	$A/B = A \times \text{inv}(B)$ $A \setminus B = \text{inv}(A) \times B$
sum of array	$\text{sum}()$	max length	$\text{length}()$	size of array	$\text{size}(\text{column}, \text{row})$
$C = A \times B$ $A = \begin{pmatrix} A11 & A12 \\ A21 & A22 \\ A31 & A32 \end{pmatrix}$ $B = \begin{pmatrix} B11 & B12 & B13 \\ B21 & B22 & B23 \end{pmatrix}$ $C = \begin{pmatrix} (A11 \times B11) + (A12 \times B21) & (A11 \times B12) + (A12 \times B22) & (A11 \times B13) + (A12 \times B23) \\ (A21 \times B11) + (A22 \times B21) & (A21 \times B12) + (A22 \times B22) & (A21 \times B13) + (A22 \times B23) \\ (A31 \times B11) + (A32 \times B21) & (A31 \times B12) + (A32 \times B22) & (A31 \times B13) + (A32 \times B23) \end{pmatrix}$					

Solve linear equations

$3x + 2y - z = 5$ $5y + 2z = 2$ $2x + 3y - 8z = 9$	$Ax = B$ $x = A^{-1}B$	$A = [3 \ 2 \ -1 ; 0 \ 5 \ 2 ; 2 \ 3 \ -8]$ $B = [5 \ 2 \ 9]'$	$x = \text{inv}(A) \times B$ or $A \setminus B$
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Statement, loops and functions

if statement	switch statement	for loop	while loop	function
$\text{if}(\text{condition1})$ $\dots \text{code} \dots$ $\text{elseif}(\text{condition2})$ $\dots \text{code} \dots$ $\text{elseif}(\text{condition3})$ $\dots \text{code} \dots$ else $\dots \text{code} \dots$ end	switch expression $\text{case}(\text{value1})$ $\dots \text{code} \dots$ $\text{case}(\text{value2})$ $\dots \text{code} \dots$ otherwise $\dots \text{code} \dots$ end	$\text{for start} : \text{end}$ $\dots \text{code} \dots$ end	while condition $\dots \text{code} \dots$ end	$\text{function}[\text{out1}, \text{out2}] = \text{functionName}(\text{input1}, \text{input2})$ $\dots \text{code} \dots$ End call function $\text{variable} = \text{functionName}(\text{input1}, \text{input2})$

jump out the loop	break	directly into next loop	continue	back before loop	return
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Plotting

Label line	<code>legend()</code>	add title	<code>title()</code>	adjust axis	<code>axis square</code> or <code>equal</code> or <code>tight</code>
label axis	<code>x/y/zlabel(' ')</code>				
multiple curve	<code>...plot...</code> <code>hold on</code> <code>...plot...</code> <code>hold off</code>	multiple graph for $m \times n$ array at position p	<code>subplot(m,n,p)</code> <code>...plot...</code>	set graph	<code>set(h, name, value)</code>

2D plots	<code>plot(x,y,'symbols, linestyle, colours')</code>	3D plots	<code>ezsurf(' ')</code>
	<code>ezplot(' ')</code>		<code>ezmesh(' ')</code>

Complex number

$a+bi$	<code>=a+bi</code> <code>=complex(a,b)</code>	real parts	<code>=real()</code>	imaginary parts	<code>imag()</code>
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Structures and cell arrays

structure	record						<code>record(1).name='John'</code>
	record(1)		record(2)		record(3)		<code>record(1).age=15</code>
	.name	John	.name	James	.name	Jack	<code>record(1).grade=[100 30;25 60]</code>
	.age	15	.age	20	.age	18	<code>record(2).name='James'</code>
	.grade	$\begin{pmatrix} 100 & 30 \\ 25 & 60 \end{pmatrix}$	.grade	$\begin{pmatrix} 90 & 85 \\ 75 & 65 \end{pmatrix}$	.grade	$\begin{pmatrix} 20 & 30 \\ 23 & 15 \end{pmatrix}$	<code>...</code> <code>record</code>

cell	1	'text'		<code>variable={1,'text';[100 30;25 60],[2,'text';[100 30;25 60],2+3i}</code>
	$\begin{pmatrix} 100 & 30 \\ 25 & 60 \end{pmatrix}$	2	'text'	OR <code>variable(1,1)=1</code> <code>variable(1,2)='text'</code> <code>...</code>